Low Error, Wrong Response: Causal Blindness in Multivariate Time-Series Forecasting

Anonymous Author(s)
Affiliation
Address
email

Abstract

Multivariate time-series forecasting models are usually selected by observational errors such as MSE and MAE. This practice assumes that accurate prediction implies that the model has learned useful temporal and cross-channel dependencies. We show that this assumption can fail in a controlled but revealing way. We identify *causal blindness*: a forecaster can obtain low observational error while producing an almost zero response to an intervention on the true driver of the target. We construct a multivariate forecasting benchmark with one cause, one target, and nineteen distractors, where the mechanism is $Y_t = 2C_{t-1} + \epsilon_t$ and the horizon-1 counterfactual response is analytically known. Across DLinear, PatchTST, iTransformer, Crossformer, TimeMixer, and DUET, a standardized intervention whose expected response is about 5.0 often yields predicted responses close to zero, despite strong sensitivity to the target's own history. We introduce intervention-aware diagnostics, including interventional response error and response slope. A functional sensitivity analysis shows that the learned predictors are locally far more sensitive to target history than to the true last-step driver, providing an operational explanation for the failure. We propose Randomized Intervention-Response Regularization (RIR), a lightweight training objective that regularizes the forecast response to known interventions. RIR repairs response curves for both iTransformer and DUET-Mix, and an augmented ETTh1 diagnostic suggests that response alignment need not degrade the ordinary MSE/MAE of original benchmark channels.

1 Introduction

Multivariate long-term time-series forecasting (MLTSF) has advanced rapidly through architectures based on linear projections, patching, transformers, inverted attention, cross-dimensional attention, decomposition, multiscale mixing, and dual clustering [Zeng et al., 2023, Nie et al., 2023, Liu et al., 2024, Zhang and Yan, 2023, Wang et al., 2024, Qiu et al., 2025]. Despite architectural differences, most models are selected and compared using observational forecasting errors such as MSE and MAE. This evaluation practice implicitly assumes that if a model predicts a target well, then it has learned the relevant temporal and cross-channel dependencies.

We argue that this assumption is incomplete. In many scientific, industrial, and operational settings, a forecaster is not only expected to extrapolate smooth curves; it is expected to respond correctly when a driver changes. For example, load should respond to upstream temperature or demand changes, downstream traffic should respond to upstream flow, and physical targets should respond to their known input drivers. A model that obtains low MSE by relying on target autocorrelation may still be unreliable under interventions, distribution shifts, or policy changes.

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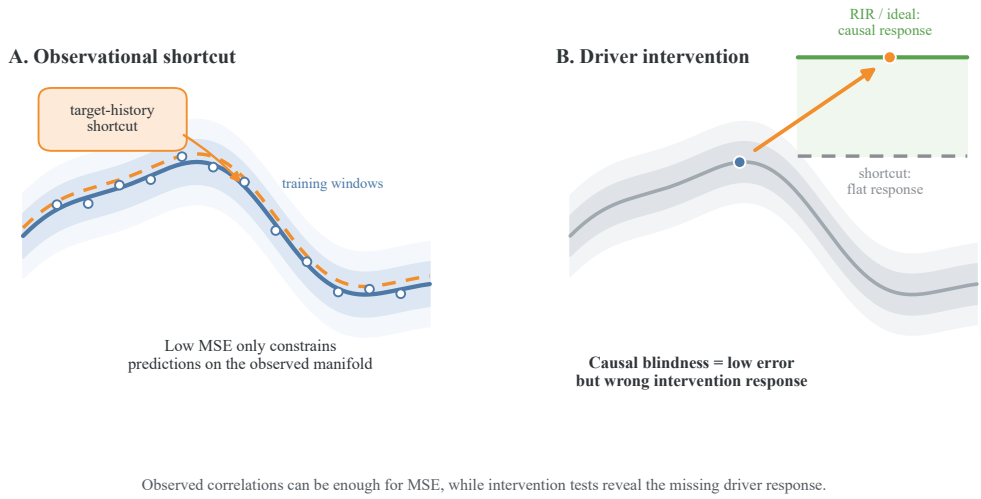

Figure 1: Causal blindness. Observational training can fit the data manifold with low MSE by exploiting target-history shortcuts. When the true driver is intervened on, the correct response moves off the observed manifold, but a shortcut forecaster can remain almost unchanged.

This paper studies the following question:

> *Do modern multivariate forecasters respond to the true causal driver when the driver is intervened on?*

We show that the answer can be no, even when the causal mechanism is simple and analytically known. We construct a controlled benchmark with $N = 21$ channels: one cause C, one target Y, and nineteen distractors. The target is generated by

$$Y_t = 2C_{t-1} + \epsilon_t, \tag{1}$$

with small noise. Since the mechanism is known, changing the last historical cause value has a known horizon-1 counterfactual effect on the target. This makes it possible to evaluate not only whether forecasts are accurate, but whether their *response* to a cause intervention has the correct direction and magnitude.

Our empirical finding is striking. Across DLinear, PatchTST, iTransformer, Crossformer, TimeMixer, and DUET, a standardized cause shift whose expected horizon-1 target response is approximately 5.000 leads to predicted changes near zero. In particular, a strong DUET configuration obtains low target MSE but predicts only 0.029 for an intervention whose expected response is 5.000. Meanwhile, zeroing the target history substantially changes the forecasts for all models. In other words, these models are not generally insensitive; rather, they are highly sensitive to the target's own history and weakly sensitive to the true causal driver.

We call this failure mode *causal blindness*. It differs from poor forecasting: some affected models achieve low observational error. It also differs from generic robustness failure: the problem is not merely that inputs are perturbed, but that the model's response is misaligned with an analytically known mechanism.

Our contributions are:

- We formulate causal blindness as a hidden failure mode of MLTSF models: low observational forecasting error can coexist with near-zero response to the true causal driver.
- We construct a controlled multivariate forecasting benchmark with known horizon-1 counterfactual responses, enabling direct evaluation of response correctness rather than only prediction accuracy.
- We introduce intervention-aware diagnostics: cause sensitivity, distractor sensitivity, target-history sensitivity, horizon-1 interventional response error, response slope, and sign accuracy.

- We provide a functional sensitivity diagnosis showing that low-error predictors can be locally aligned with target-history shortcuts rather than the true driver; RIR changes this local response geometry.
- We provide evidence across five representative model families and a recent DUET baseline, showing that causal blindness is not restricted to a single architecture or to channel-independent modeling.
- We instantiate a simple remedy, Randomized Intervention-Response Regularization, that regularizes the response to randomized counterfactual interventions. RIR raises the predicted response from near zero to more than 80% of the analytically expected effect on iTransformer and DUET-Mix, and it recovers near-linear response curves across positive and negative intervention magnitudes.
- We include an augmented ETTh1 diagnostic showing that response alignment improves a semi-synthetic causal target without degrading the MSE/MAE of the original `OT` channel across three seeds.

2 Related Work

Modern MLTSF models include transformer-based sequence models [Zhou et al., 2021, Wu et al., 2021, Zhou et al., 2022], channel-independent patch models [Nie et al., 2023], linear baselines [Zeng et al., 2023], inverted and cross-dimensional transformers [Liu et al., 2024, Zhang and Yan, 2023], multiscale mixing [Wang et al., 2024], and dual clustering [Qiu et al., 2025]. These works primarily optimize observational MSE/MAE. Our contribution is not another forecasting backbone; it is an intervention-aware evaluation and training protocol showing that cross-channel access does not imply correct causal response.

Our work connects to shortcut learning [Geirhos et al., 2020], invariant and causal representation learning [Arjovsky et al., 2019, Schölkopf et al., 2021], and classical intervention-based causal reasoning [Granger, 1969, Pearl, 2009, Peters et al., 2017]. Recent forecasting work has begun to address causal evaluation and counterfactual reasoning [Crasson et al., 2024, Yan and Wang, 2023, Yang et al., 2025]. We differ in focus: rather than estimating treatment effects from observational regimes or generating explanations, we test whether standard MLTSF predictors have the correct numerical response to known driver interventions.

3 Problem Setup and Evaluation Metrics

Let $X_{1:T} \in \mathbb{R}^{T\times N}$ be a history window and $Z_{T+1:T+H} \in \mathbb{R}^{H\times N}$ the full future window. A forecaster f_θ is usually selected by observational error,

$$\mathcal{L}_{\text{obs}} = \frac{1}{HN} \left\| f_\theta(X_{1:T}) - Z_{T+1:T+H} \right\|_F^2 . \tag{2}$$

We ask a different question: if a target channel y is driven by a cause channel c, does the horizon-1 prediction respond correctly when c is intervened on? For an intervention a, define

$$\Delta\hat{y}_1(a) = f_\theta(X^a)_{1,y} - f_\theta(X)_{1,y}, \qquad \Delta y_1(a) = Z^a_{1,y} - Z_{1,y}. \tag{3}$$

We call a model *causally blind* when it has low observational error but a large response error $|\Delta\hat{y}_1(a) - \Delta y_1(a)|$ for a known driver intervention.

Scope of causal evaluation. Unlike observational forecasting, interventional evaluation requires ground-truth counterfactual responses. Standard benchmarks such as Traffic, Electricity, and raw ETT provide observed futures but no *do*-labels for channel interventions, so raw benchmark accuracy alone is not causal evidence. We therefore use settings where the response is known by construction: a controlled benchmark for causal evidence and an augmented real-world benchmark for side-effect and realism diagnostics.

Controlled benchmark. We construct a 21-channel benchmark with one cause C, one target Y, and 19 distractors. The structural equation is

$$Y_t = 2C_{t-1} + \epsilon_t, \tag{4}$$

Table 1: Main causal-blindness result. The expected horizon-1 response to $C_T \leftarrow C_T + 5$ is 5.000384. "Pred. $|\Delta|$" reports the absolute mean target response; slope is computed from signed responses.

Model	Target MSE	Target MAE	Pred. $\lvert\Delta\rvert$	H1 IRE	Slope	Target-zero MPD
DLinear	0.5454	0.6220	0.000	5.000	0.000	0.648
PatchTST	0.2960	0.4272	0.000	5.000	0.000	0.799
iTransformer	0.0608	0.1909	0.032	4.994	0.001	0.870
Crossformer	0.0157	0.0994	0.001	5.000	0.000	0.884
TimeMixer	0.0109	0.0831	0.000	5.000	0.000	0.838
DUET-Mix	0.0245	0.1262	0.053	4.998	0.000	0.789

with small Gaussian noise. The input length and prediction length are both 96. All channels are standardized once using training-set statistics; interventions are then applied in standardized input space. The final channel order is $C, D_1, \ldots, D_{19}, Y$.

Our causal evaluation uses a last-step cause intervention,

$$C'_T = C_T + \delta, \qquad C'_t = C_t \text{ for } t < T. \tag{5}$$

Since Eq. (4) implies $Y_{T+1} = 2C_T + \epsilon_{T+1}$, the horizon-1 counterfactual response is exact and does not require assumptions about future cause trajectories. In standardized coordinates,

$$\Delta y^{\text{std}}_{T+1} = 2\frac{\sigma_C}{\sigma_Y}\Delta C^{\text{std}}_T, \tag{6}$$

and in our benchmark $2\sigma_C/\sigma_Y = 1.000076888$, so $\delta = 5$ implies an expected response of 5.000384. We report horizon-1 interventional response error (IRE), response slope, and mean prediction difference (MPD). Full metric definitions, all-history sensitivity tests, and normalization caveats are in Appendix A.

4 Causal Blindness: Diagnosis and Why it Happens

We evaluate DLinear, PatchTST, iTransformer, Crossformer, TimeMixer, and DUET. The architectures span linear decomposition, channel-independent patching, variable-token attention, cross-dimensional attention, multiscale mixing, and dual temporal-channel clustering. Table 1 shows the central failure: models with low target MSE can have almost zero response to the true driver. TimeMixer has the lowest MSE among the Time-Series-Library models but zero measured response; DUET-Mix also obtains low error while predicting only 0.053 for an expected response of 5.000.

This result is not explained away by preprocessing. A correctly specified ARX baseline using C_T recovers a response slope of 0.9995 under the same standardization, while a target-only AR baseline remains non-responsive. Nor is the effect only a large out-of-distribution shift: for $\delta \in \{\pm 0.1, \pm 0.25, \pm 0.5, \pm 1\}$, iTransformer has small-delta slope -0.022, PatchTST and TimeMixer have slope 0, and ARX remains calibrated (0.9995). Table 2 summarizes these checks. The ARX result is important because it uses the same standardization and evaluation code as the deep forecasters; therefore a near-zero response is not forced by the data pipeline. The small-delta result is equally important because it shows that causal blindness is already visible in the local perturbation regime, not only under the visually striking $\delta = 5$ intervention. Appendix C further varies the look-back window for DUET-Mix and finds near-zero response slopes from $T = 48$ to $T = 336$.

The failure patterns are informative. Strict channel-independent predictors can be structurally blind because the target prediction path cannot use another channel. Channel-mixing models have access to the cause, but access is not alignment: observational training can still favor target-history shortcuts. Crossformer also responds more to distractor shifts than to the true cause in broad sensitivity tests, suggesting that cross-channel interaction is not necessarily causal interaction.

Why can low MSE coexist with wrong response? Observational error constrains a predictor's values on the observed data manifold, but not its response to interventions. Suppose $Y_{T+1} = g(C_T) + \epsilon$ and the target history $S = X_{1:T,Y}$ predicts the driver well under the observational distribution: $C_T = q(S) + \nu$ with $\mathbb{E}\nu^2 \leq \eta^2$. If g is L-Lipschitz, the shortcut $h(S) = g(q(S))$

Table 2: Sanity checks for the response evaluation. ARX confirms that the same preprocessing and evaluation pipeline can recover the correct response. Small-delta tests use $\delta \in \{\pm 0.1, \pm 0.25, \pm 0.5, \pm 1\}$ and show that near-zero response is not merely a large-shift artifact.

Check	Model	H1/Target MSE	Small-delta slope	Full-sweep slope	Mean IRE
Pipeline	AR(Y_T)	0.0183	0.0000	0.0000	–
Pipeline	ARX(C_T)	0.0010	**0.9995**	**0.9995**	0.0002
Local response	iTransformer	0.0608	-0.0220	-0.0008	0.4737
Local response	PatchTST	0.2960	0.0000	0.0000	0.4625
Local response	TimeMixer	0.0109	0.0000	0.0000	0.4625

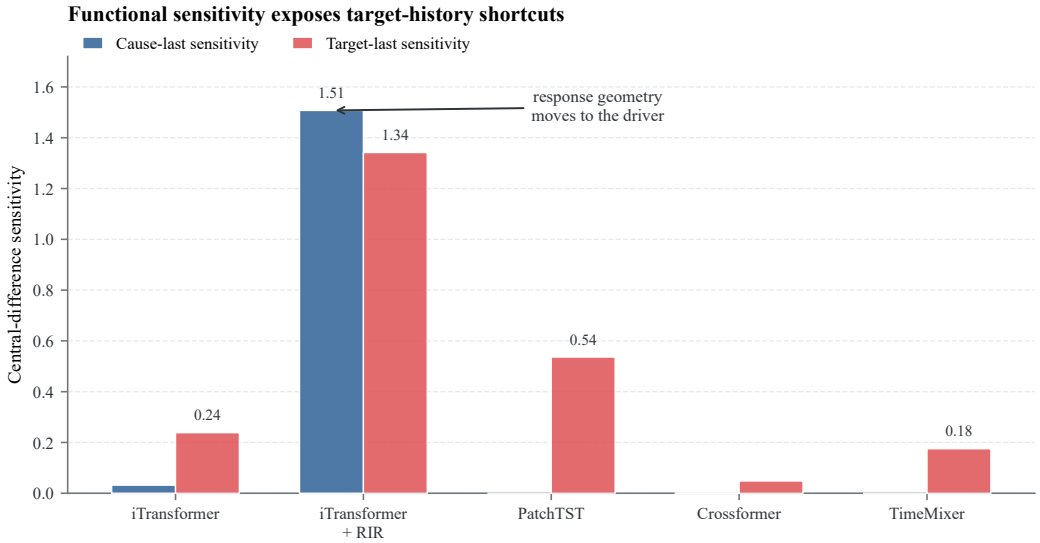

Figure 2: Functional input sensitivity of the horizon-1 target prediction, measured by central differences with $\tau = 0.05$. MSE-trained models are mostly aligned with target-history shortcuts; RIR restores cause sensitivity.

satisfies

$$\mathbb{E}\left(Y_{T+1} - h(S)\right)^2 \leq L^2\eta^2 + \mathbb{E}\epsilon^2 \tag{7}$$

when the structural noise is mean-zero and independent of ν. Thus a predictor that ignores C_T can be accurate whenever S predicts C_T on the training manifold. Under the intervention $C_T \leftarrow C_T + \delta$, however, the shortcut has zero response while the structural response is $g(C_T + \delta) - g(C_T)$.

We test this explanation directly using functional sensitivity. With $\tau = 0.05$, we measure

$$S(a) = \mathbb{E}\left|\frac{f_\theta(X + \tau a)_{1,Y} - f_\theta(X - \tau a)_{1,Y}}{2\tau}\right|, \tag{8}$$

where a perturbs either the last cause value or the last target-history value. Figure 2 shows that MSE-trained models are locally much more sensitive to target history than to the true last-step driver. RIR reverses this imbalance for iTransformer.

5 Proposed Method: Randomized Intervention-Response Regularization

RIR is a lightweight response-supervision objective for settings where domain knowledge, a simulator, or benchmark construction provides a known driver and horizon-1 response. During training, we sample $\delta_i = s_i m_i$, where $s_i \in \{-1, 1\}$ is uniform and $m_i \sim \mathrm{Uniform}(\rho\Delta_{\max}, \Delta_{\max})$. We intervene only on the last historical driver value and define $\Delta y_i^\star = (2\sigma_C/\sigma_Y)\delta_i$ for the synthetic benchmark.

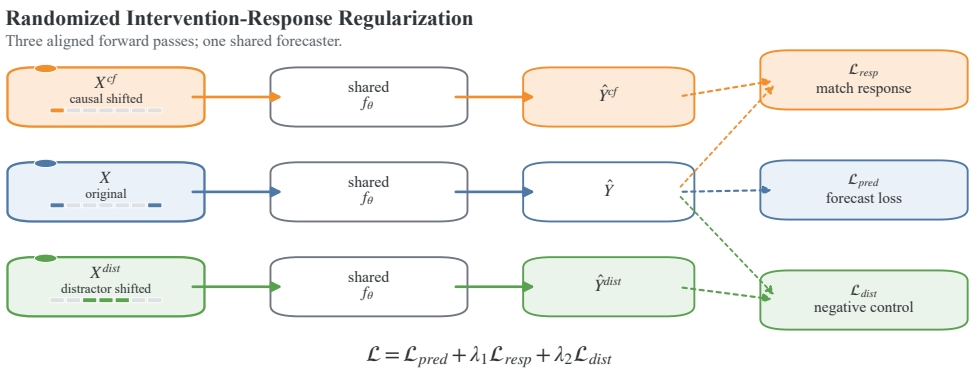

Figure 3: RIR training workflow. The same forecaster is evaluated on the original input, a last-step causal intervention, and a distractor intervention. The response loss aligns the known driver response, while the distractor loss discourages indiscriminate sensitivity.

The response loss is sliced to the horizon-1 target:

$$\mathcal{L}_{\text{resp}} = \frac{1}{B} \sum_{i=1}^{B} \left[f_\theta(X_i^{\text{cf}})_{1,Y} - f_\theta(X_i)_{1,Y} - \Delta y_i^\star \right]^2 . \tag{9}$$

To avoid indiscriminate reactivity, we also perturb last-step distractor values and penalize target response:

$$\mathcal{L}_{\text{dist}} = \frac{1}{B} \sum_{i=1}^{B} \left[f_\theta(X_i^{\text{dist}})_{1,Y} - f_\theta(X_i)_{1,Y} \right]^2 . \tag{10}$$

The training loss is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_1 \mathcal{L}_{\text{resp}} + \lambda_2 \mathcal{L}_{\text{dist}}. \tag{11}$$

RIR adds no parameters, but uses three forward passes per step: original, causal intervention, and distractor intervention.

What supervision does RIR require? RIR is not intended to replace causal discovery. It is useful when a practitioner has a trusted response label for at least one driver-target pair, for example from a physical simulator, a controlled intervention, an engineering specification, or a benchmark construction. The label need not specify a full causal graph or every future horizon. In our experiments, it specifies only the horizon-1 response of a single target to a last-step driver intervention. This limited supervision is enough to change the local response geometry of the forecaster. In domains where only partial knowledge is available, the same objective can be applied sparsely: supervise known driver-target pairs, leave unknown pairs unconstrained, and use distractor or negative-control variables when domain knowledge indicates zero response.

Why include the distractor term? Without $\mathcal{L}_{\text{dist}}$, response supervision can teach a model to be reactive without teaching it where to be reactive. In our ablations (Appendix D), a response-only loss repairs the cause response but also creates large false response to last-step distractor interventions. The distractor term acts as a local negative control: it asks the model to change under the known driver while remaining stable under variables whose intervention should not affect the target. This distinction is central to our framing. The goal is not generic sensitivity, but selective response aligned with known mechanisms.

6 Experiments

6.1 Synthetic Benchmark

Table 3 shows that RIR restores most of the missing response in both iTransformer and DUET-Mix. Fine-tuning trades some observational MSE for response alignment, while DUET-Mix keeps target

Table 3: RIR repair results. DUET-Mix rows report mean ± standard deviation over three seeds; iTransformer rows are single-seed diagnostics.

Model	Target MSE	Target MAE	Pred. Δ	H1 IRE	Slope
iTransformer	0.0608	0.1909	0.032	4.994	0.001
iTransformer + RIR FT01	0.1054	0.2553	4.149	0.871	0.830
iTransformer + RIR FT03	0.1465	0.3020	4.205	0.816	0.841
DUET-Mix	0.0237±0.0017	0.1245±0.0043	0.054±0.002	4.982±0.017	0.004±0.003
DUET-Mix + RIR	0.0308±0.0066	0.1224±0.0084	4.259±0.086	0.744±0.091	0.852±0.017

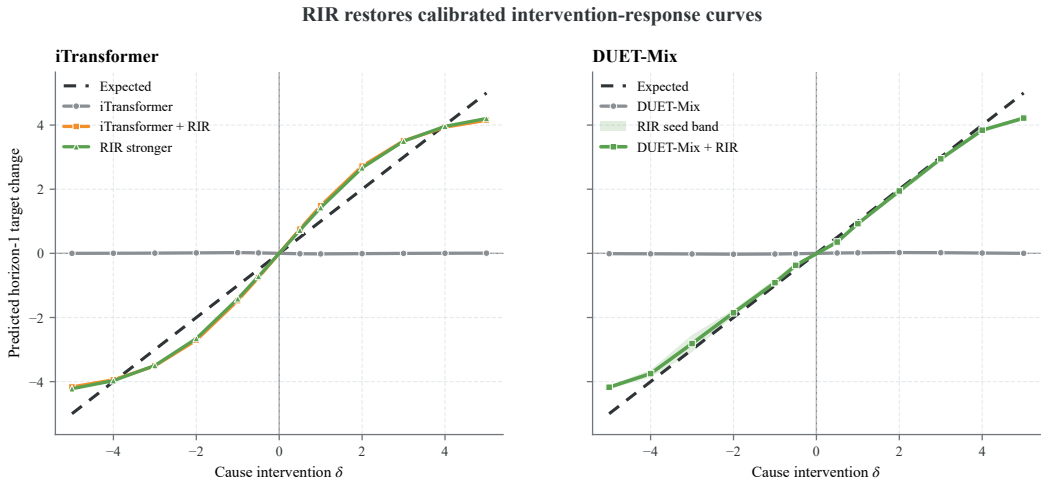

Figure 4: Delta-response curves for two channel-mixing architectures. The dashed line is the expected response. MSE-trained models are nearly flat, while RIR tracks the intervention-response function across positive and negative shifts.

MAE comparable to the baseline. Figure 4 shows that the repair generalizes beyond the single $+5$ intervention. The MSE-only iTransformer and DUET-Mix response curves remain nearly flat, while RIR recovers approximately linear response curves across positive and negative shifts.

6.2 Semi-Synthetic Real Data: Augmented ETTh1

The controlled benchmark provides exact causal labels, but it is synthetic. We therefore construct an augmented ETTh1 diagnostic: retain all original ETTh1 channels, including `OT`, and append

$$Y_t^{\mathrm{rir}} = 2 \cdot \mathrm{HUFL}_{t-1} + s(t) + \epsilon_t, \tag{12}$$

where $\epsilon_t \sim \mathcal{N}(0, 0.05^2)$ and $s(t) = 2[\sin(2\pi h/24) + 0.5\cos(2\pi h/24) + 0.5\sin(2\pi d/365.25)]$ for hour h and day-of-year d. RIR is applied only to the appended target. Thus raw `OT` accuracy is a side-effect diagnostic rather than causal evidence. Table 4 shows that RIR recovers the response curve of the semi-synthetic target without degrading raw `OT` MSE/MAE. Appendix F repeats the same augmented diagnostic on ETTh2 and finds the same response collapse and repair, with modest dataset-dependent side effects on raw `OT`.

This experiment should be interpreted as a side-effect diagnostic rather than as a claim that raw ETTh1 contains observable causal intervention labels. The appended target gives us a known response while the original channels preserve realistic seasonality, scaling, correlations, and timestamp structure. Because `OT` is trained and evaluated normally but never receives a response label, its MSE/MAE tests whether adding response supervision for one channel causes collateral damage to an ordinary benchmark target. The result suggests that response alignment can be added locally: it improves the channel for which response knowledge is available without necessarily degrading unrelated channels optimized under standard forecasting loss.

Table 4: Augmented ETTh1 side-effect diagnostic at prediction length 96, reported as mean $\pm$ standard deviation over three seeds.

Variant	All MSE	Raw OT MSE	Raw OT MAE	Semi MSE	Curve slope	Curve IRE
Baseline	0.4267 ± 0.0022	0.0559 ± 0.0002	0.1828 ± 0.0004	0.7144 ± 0.0039	0.0226 ± 0.0023	2.4149 ± 0.0056
+ RIR	0.4265 ± 0.0043	0.0557 ± 0.0005	0.1817 ± 0.0007	0.7078 ± 0.0083	0.9292 ± 0.0121	0.3317 ± 0.0057

7 Discussion and Conclusion

Our cleanest evidence uses a linear one-lag synthetic mechanism. This choice makes the horizon-1 counterfactual response unambiguous, but it does not by itself prove the same magnitude of blindness for nonlinear, multi-lag, or confounded mechanisms. The augmented ETT diagnostics improve external validity but remain semi-synthetic; raw Traffic, Electricity, or ETT benchmarks would test observational accuracy rather than causal response unless augmented with known response labels. Some core tables still use single-seed diagnostics or historical checkpoints, while DUET-Mix and the augmented ETT diagnostics are already evaluated over three seeds. A final submission should continue with unified multi-seed reruns and a supplementary reproducibility package. Finally, RIR assumes known driver-response labels; it is not a general causal discovery method, and misspecified response labels could align a model to an incorrect domain belief.

The broader implication is evaluative. Current MLTSF leaderboards reward models for interpolation on observational test windows. That protocol is appropriate when the deployment distribution is stable and no controllable driver is changed. It is insufficient, however, for settings where forecasts are used under policy changes, control actions, physical inputs, or upstream interventions. In those settings, a forecasting benchmark should include at least a small number of response-aware unit tests: known driver interventions, negative-control distractor interventions, and local response slopes. Such tests do not require solving general causality, but they prevent a model from looking reliable solely because it follows target-history autocorrelation.

We identify causal blindness as a hidden failure mode of multivariate time-series forecasting: low MSE/MAE can coexist with near-zero response to the true driver. Across representative architectures, including DUET-Mix, the learned functions are often locally aligned with target-history shortcuts rather than the causal input. Intervention-aware diagnostics expose this gap, and RIR shows that known response labels can steer existing forecasters toward the right response without necessarily damaging ordinary benchmark accuracy.

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A Benchmark, Interventions, and Metrics

Data and channel map. The controlled benchmark contains one cause C, one target Y, and 19 distractors. We use 10,000 time steps, split into 70% training, 10% validation, and 20% test. The history length and prediction horizon are both 96. Variables are standardized using training-set statistics and then reordered as channel 0 = C, channels 1–19 = distractors, and channel 20 = Y.

All-history sensitivity interventions. For broad sensitivity diagnostics, we perturb entire historical channels in standardized input space:

$$\text{cause shift}: \quad C'_{1:T} = C_{1:T} + 5, \qquad \text{cause zero}: \quad C'_{1:T} = 0, \tag{13}$$

$$\text{cause flip}: \quad C'_{1:T} = -C_{1:T}, \qquad \text{distractor shift}: \quad D'_{1:19,1:T} = D_{1:19,1:T} + 5, \tag{14}$$

$$\text{target zero}: \quad Y'_{1:T} = 0. \tag{15}$$

All-history additive shifts can be attenuated by per-instance normalization inside some forecasting backbones, so the main causal evidence in the paper uses last-step interventions with analytic horizon-1 labels.

Metrics. For an intervention a, mean prediction difference is

$$\mathrm{MPD}(a) = \mathbb{E}\left[|f_\theta(X^a)_{1,Y} - f_\theta(X)_{1,Y}|\right]. \tag{16}$$

When the counterfactual label is known, horizon-1 interventional response error is

$$\mathrm{IRE}_{H1}(a) = \mathbb{E}\left[\left|(f_\theta(X^a)_{1,Y} - f_\theta(X)_{1,Y}) - \left(Z^a_{1,Y} - Z_{1,Y}\right)\right|\right]. \tag{17}$$

The signed response slope is

$$\text{slope} = \frac{\sum_i \Delta\hat{y}_{1,i}\Delta y_{1,i}}{\sum_i (\Delta y_{1,i})^2}. \tag{18}$$

For response curves, we also report the mean signed response ratio over nonzero intervention magnitudes.

B Additional Causal-Blindness Diagnostics

Table 5: All-history sensitivity evaluation. This diagnostic is useful for identifying target-history reliance, but all-history additive shifts can interact with model-internal normalization.

Model	Target MSE	Target MAE	Cause MPD	Distractor MPD	Target-zero MPD
DLinear	0.545448	0.622045	0.000000	0.000000	0.562089
PatchTST	0.296033	0.427234	0.000000	0.000000	0.730165
iTransformer	0.060812	0.190868	0.000002	0.000006	0.840644
Crossformer	0.015730	0.099358	0.003551	0.034936	0.878338
TimeMixer	0.010938	0.083087	0.000000	0.000000	0.857566

Table 6: Additional DUET evaluation. All variants obtain low observational error but near-zero horizon-1 response to the true cause.

Variant	Loss	Target MSE	Target MAE	Pred. $\lvert\Delta\rvert$	H1 IRE	Slope
DUET-CI	MSE	0.018961	0.111400	0.029323	4.998435	0.000390
DUET-Mix	MSE	0.024529	0.126233	0.052558	4.998288	0.000419
DUET-CI	MAE	0.027745	0.134924	0.040672	4.974961	0.005084

Table 7: Linear sanity baselines under the same preprocessing. A correctly specified ARX baseline recovers the expected response, confirming that the evaluation pipeline can represent the causal response.

Model	H1 MSE	H1 MAE	Cause coef.	Small slope	Full slope
AR(Y_T)	0.0183	0.1096	0.0000	0.0000	0.0000
ARX(C_T)	0.0010	0.0251	0.9996	0.9995	0.9995
ARX(C_T, Y_T)	0.0010	0.0251	1.0037	1.0036	1.0036
VARX(X_T)	0.0010	0.0253	1.0040	1.0039	1.0039

Table 8: Small-delta response evaluation using $\delta \in \{\pm 0.1, \pm 0.25, \pm 0.5, \pm 1\}$. The flat response appears in the local perturbation regime, not only under a large ± 5 shift.

Model	Small-delta slope	Mean IRE	Mean ratio
ARX(C_T)	0.9995	0.0002	0.9995
iTransformer	-0.0220	0.4737	-0.0273
iTransformer + RIR FT01	1.4769	0.2304	1.4955
PatchTST	0.0000	0.4625	0.0000
Crossformer	0.0000	0.4625	0.0000
TimeMixer	0.0000	0.4625	0.0000

C Look-Back Window Robustness

Table 9 varies the history length for DUET-Mix while keeping the prediction length, mechanism, and last-step intervention fixed. The response collapse persists across all tested windows. Increasing the look-back window from 96 to 192 lowers target MSE, but the response slope remains essentially zero. This supports the interpretation that causal blindness is not an artifact of the particular $T = 96$ setting.

Table 9: Look-back window robustness for DUET-Mix baseline on the controlled synthetic benchmark. Prediction length is fixed to 96. The expected H1 response to $C_T \leftarrow C_T + 5$ is approximately 5.000.

T	Target MSE	Target MAE	Pred. H1	H1 IRE	H1 Slope	Target-zero MPD	Curve slope
48	0.3329	0.4645	0.0883	4.9760	0.0049	0.7988	0.0048
96	0.0248	0.1278	0.0568	4.9811	0.0038	0.7776	0.0007
192	0.0157	0.1019	0.0226	5.0021	-0.0003	0.8024	-0.0012
336	0.0205	0.1155	0.0244	4.9979	0.0005	0.7812	-0.0013

D Additional RIR Results

For iTransformer synthetic experiments, RIR uses $\Delta_{\max} = 5$ and $\rho = 0.5$. FT01 and FT03 initialize from the MSE-only checkpoint and use $(\lambda_1, \lambda_2) = (0.1, 0.01)$ and $(0.3, 0.03)$, respectively. The scratch variant uses $(1.0, 0.1)$. DUET and ETTh1 augmented runs use $\rho = 0.1$.

The DUET-CI + RIR diagnostic should not be interpreted as evidence that a strictly channel-independent function can use another channel at inference. The inspected DUET implementation contains shared training components and auxiliary mechanisms, so response supervision can still alter the learned predictor. We therefore use DLinear and PatchTST as stricter channel-independent negative controls.

E Augmented ETTh1 Details

For ETTh1 augmented runs, RIR is applied only to the appended semi-synthetic target and uses $(\lambda_1, \lambda_2) = (0.05, 0.005)$. The original `OT` channel is never given a response label.

Table 10: iTransformer RIR variants.

Model	Target MSE	Target MAE	Pred. Δ	H1 IRE	Slope
iTransformer	0.0608	0.1909	0.032	4.994	0.001
+ RIR FT01	0.1054	0.2553	4.149	0.871	0.830
+ RIR FT03	0.1465	0.3020	4.205	0.816	0.841
+ RIR scratch	0.5308	0.5910	4.794	0.243	0.959

Table 11: RIR ablation on iTransformer. The distractor-stability term reduces false response to last-step distractor interventions.

Model	Target MSE	Cause response	H1 IRE	Slope	Last-dist. false resp.
iTransformer	0.0608	0.032	4.994	0.001	0.200
MSE fine-tune	0.0826	0.062	4.987	0.003	0.262
+ $\mathcal{L}_{\text{resp}}$	0.1095	4.277	0.755	0.855	5.929
+ $\mathcal{L}_{\text{resp}} + \mathcal{L}_{\text{dist}}$	0.1054	4.149	0.871	0.830	0.672

Table 12: Delta-response curve evaluation on iTransformer variants.

Model	Curve slope	Curve corr.	Mean IRE	Mean ratio
iTransformer	-0.0008	-0.2276	2.5913	-0.0090
MSE fine-tune	-0.0001	-0.0126	2.5906	-0.0103
+ $\mathcal{L}_{\text{resp}}$	0.9969	0.9865	0.5112	1.2178
+ $\mathcal{L}_{\text{resp}} + \mathcal{L}_{\text{dist}}$	0.9836	0.9839	0.5447	1.2216
+ RIR FT03	0.9867	0.9862	0.5096	1.1983

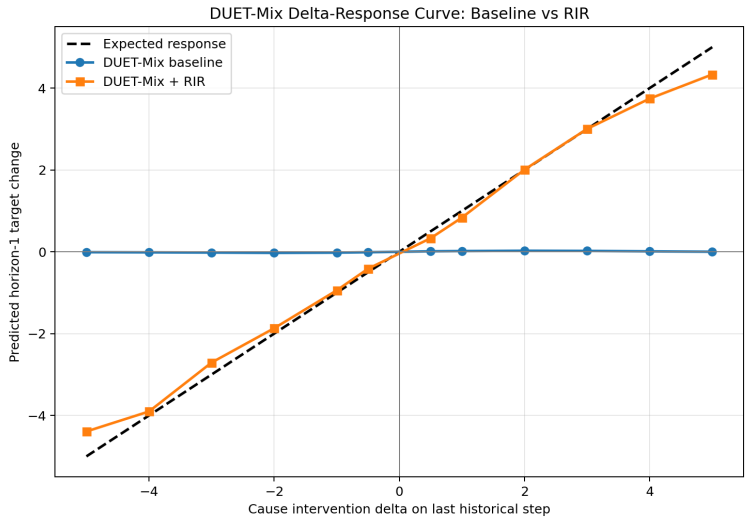

Figure 5: DUET-Mix delta-response curves. The baseline is nearly flat, while RIR tracks the analytically expected intervention-response function.

Table 13: Full ETTh1 augmented side-effect diagnostic. Prediction length 96 uses three seeds; prediction length 192 is a one-seed probe.

Pred. len	Variant	All MSE	Raw OT MSE	Raw OT MAE	Semi MSE	Curve slope	Curve IRE
96	Baseline	0.4267±0.0022	0.0559±0.0002	0.1828±0.0004	0.7144±0.0039	0.0226±0.0023	2.4149±0.0056
96	+ RIR	0.4265±0.0043	0.0557±0.0005	0.1817±0.0007	0.7078±0.0083	0.9292±0.0121	0.3317±0.0057
192	Baseline	0.4906	0.0728	0.2081	0.8375	0.0147	2.4349
192	+ RIR	0.4838	0.0728	0.2071	0.8220	0.9531	0.3082

F Augmented ETTh2 Generalization Diagnostic

To check whether the augmented-real diagnostic is specific to ETTh1, we repeat the same construction on ETTh2. We retain the original ETTh2 channels, append $Y_t^{\mathrm{rir}} = 2 \cdot \mathrm{HUFL}_{t-1} + s(t) + \epsilon_t$, and apply RIR only to the appended target. Table 14 shows that the baseline response again collapses near zero, while RIR recovers an almost calibrated response curve. Unlike the ETTh1 result, raw OT error is modestly higher under RIR on ETTh2. We therefore treat this experiment as a generalization diagnostic for response repair, not as evidence that response supervision is always free of collateral effects on unrelated raw targets.

Table 14: Augmented ETTh2 diagnostic at prediction length 96, reported as mean $\pm$ standard deviation over three seeds. RIR repairs the appended target's response curve; raw OT metrics are side-effect diagnostics and are not causal evidence.

Variant	All MSE	Raw OT MSE	Raw OT MAE	Semi MSE	Curve slope	Curve IRE	Curve ratio
Baseline	0.3217 ± 0.0021	0.1475 ± 0.0014	0.2959 ± 0.0014	0.4788 ± 0.0050	0.0083 ± 0.0028	2.5959 ± 0.0077	0.0100 ± 0.0032
+ RIR	0.3245 ± 0.0061	0.1607 ± 0.0049	0.3095 ± 0.0055	0.4710 ± 0.0069	1.0033 ± 0.0215	0.5288 ± 0.0151	0.8987 ± 0.0103

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